

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 8: Winds, Storms and Cyclones

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	8 — Winds, Storms and Cyclones
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This is the hardest tier: every item chains air pressure to temperature to density to wind across several steps, and the distractors are engineered from the deepest misconceptions — treating warm air as heavier or as lifted by a special force, reversing Bernoulli so fast air seems to push outward, reading the can-crush as metal folding itself rather than an inside-versus-outside pressure imbalance, flipping wind to go from low toward high pressure, swapping the heating rates of land and sea, calling the calm eye the deadliest zone or a high-pressure core, ranking a tornado above a cyclone in size, mistaking the shorter light-then-thunder gap for a retreating storm, and reducing the storm surge to ordinary rainwater. Do not write on this paper — mark answers on your answer sheet.

1. A student weighs a rigid sealed glass globe of air on a sensitive balance, then carefully heats only the trapped air inside (the glass does not melt or leak). She predicts the balance reading will fall because warm air is 'lighter.' What actually happens, and why does this expose the trap in calling warm air lighter?

- (A) The reading falls, because heating genuinely destroys some of the air's weight
- (B) The reading is unchanged, because no air entered or left the sealed globe; warm air is only 'lighter' in the sense of lower density (same amount of air spread over more space), which matters for floating, not for the total weight on a balance
- (C) The reading rises, because warming adds energy that counts as extra weight on the balance
- (D) The reading falls, because warm air pushes up on the balance pan from inside the sealed globe

2. Two identical balloons are inflated to the same size, but balloon P is filled with cold air and balloon Q with warm air, and both are then sealed and brought to the same room temperature so the air inside each settles. A student claims P must now contain more air than Q. Reasoning carefully about how the trapped air behaves as it reaches room temperature, the correct conclusion is:

- (A) Q contains more air, because warm air expands and so more of it must have been needed to fill the same balloon
- (B) P contains more air, because cold air is denser, so filling the same volume cold packs in more air; once both warm to room temperature P's denser contents make it the heavier, more tightly filled balloon
- (C) Both contain exactly the same amount of air, because the balloons are the same size
- (D) P contains less air, because cold air shrinks and leaves empty gaps inside the balloon

3. A sealed metal can of ordinary room-temperature air (no boiling, no steam) is plunged into ice water. It dents only slightly. The same can, refilled with steam by boiling water inside before sealing, is plunged into the same ice water and crumples dramatically. A student says the difference is that steam is heavier than air. Identify the real reason and the precise flaw in the student's claim:

- (A) Steam is genuinely heavier than air, so its weight pulls the can walls inward as it cools
- (B) The boiling makes the metal thinner and weaker, so only the steam-filled can is soft enough to fold
- (C) Steam condenses to a tiny drop of liquid on cooling, leaving almost no gas inside, so the inside pressure collapses far below outside; plain air merely contracts a little, so its pressure barely drops — the flaw is that the crush comes from a pressure gap, not from any weight of steam
- (D) Cold air cannot be cooled any further, so only the steam can lose pressure on contact with ice water

4. Four weather stations lie on a straight east-west line. The gaps between neighbours are equal EXCEPT the gap between the third and fourth, which is only half as wide. Their pressures are: first 1012, second 1006, third 1002, fourth 996 (same units). Using only the high-to-low rule and 'a steeper drop over a shorter distance gives faster wind,' the single fastest stretch of wind is between:

- (A) the first and second stations, because the first has the highest pressure of all and high pressure makes the fastest wind
- (B) the third and fourth stations, because their drop of 6 over only the half-width gap is the steepest gradient of any neighbouring pair
- (C) the second and third stations, because that pair sits closest to the centre of the line
- (D) the first and fourth stations, since the largest total pressure difference is between the two end stations

5. On a sunny morning a flat field and an adjacent pond receive identical sunshine. By noon the air just above the field is hotter than the air above the pond. A student is asked to chain the cause to the resulting surface wind. The fully correct chain is:

- (A) land heats faster than water, so the air over the field warms, expands, becomes less dense and rises, lowering the surface pressure over the field, so cooler denser air drifts in from over the pond toward the field
- (B) water heats faster than land, so air rises over the pond and surface wind blows from the field toward the pond
- (C) the field's warm air is heavier, so it sinks and pushes surface air outward toward the pond
- (D) both heat equally, so no pressure difference forms and the surface air stays perfectly still all day

6. A vertical hair-dryer holds a light ball steady in its upward stream. A student gently lowers a smooth card horizontally toward the ball from one side, partly blocking the airflow on that side only. The ball is observed to drift toward the card. The best explanation is:

- (A) the card slows the air on its side, raising the pressure there, which shoves the ball away toward the open side
- (B) the moving air sticks to the ball and physically drags it sideways toward whatever object is nearest
- (C) the card speeds and channels the air on its side, lowering the pressure there, so the higher still-air pressure on the open side pushes the ball toward the card
- (D) the card adds weight to the ball's near side, tipping it over toward the card

7. A glass tumbler is filled to the brim, covered with a stiff card, and inverted; the water stays. A student now repeats it on a high mountain peak where the outside air pressure is much lower, using the same full tumbler and card. Compared with doing it at sea level, on the peak the card is:

- (A) more likely to stay, because thin mountain air clings to the card more tightly
- (B) more likely to fall, because the lower outside air pressure on the peak pushes up on the card less strongly, so it is closer to being overcome by the unchanged weight of the water
- (C) exactly as secure as at sea level, because air pressure is the same everywhere on Earth
- (D) guaranteed to stay, because water simply cannot fall when it is upside down

8. A cyclone strengthening over a warm sea drifts so that one half passes over a long, narrow strip of land while the other half stays over open warm water. Reasoning about what feeds the storm, the most likely effect on the cyclone over the next several hours is that it:

- (A) weakens overall, because the half over land is cut off from the warm moist sea air that supplies heat and vapour, so the self-feeding cycle is partly starved
- (B) strengthens, because crossing land gives the storm a solid surface to spin against and gain speed
- (C) is unaffected, because once a cyclone forms it carries its own energy and no longer needs the sea
- (D) splits permanently into two separate storms of equal strength, one over land and one over water

9. A student lists the diameters of four whirling systems: a single thunderstorm cell about 10 km across, a tornado about 0.3 km across, a full cyclone about 500 km across, and a cyclone's eye about 30 km across. She must order them from LARGEST to SMALLEST. The correct order is:

- (A) the tornado, then the full cyclone, then the eye, then the thunderstorm cell
- (B) the full cyclone, then its eye, then the thunderstorm cell, then the tornado
- (C) the full cyclone, then the thunderstorm cell, then the eye, then the tornado
- (D) the eye, then the full cyclone, then the tornado, then the thunderstorm cell

10. During a thunderstorm a child first counts about 5 seconds between a flash and its thunder, and a little later counts about 5 seconds again, then suddenly about 2 seconds. Both flash and thunder are made at the same instant. What does the steady-then-shrinking pattern tell her?

- (A) the storm moved away, because a smaller gap is produced by a more distant storm
- (B) the lightning grew dimmer, which is what shrinks the gap regardless of distance
- (C) the storm held its distance for a while and then moved noticeably closer, since a shorter gap means the sound had less distance to cover before reaching her
- (D) the speed of sound suddenly rose during the storm, closing the gap while the storm stayed put

11. A teacher connects a sealed flask of air by a thin tube to a beaker of coloured water, the tube dipping below the surface. She warms the flask and bubbles escape into the water; she cools it and water climbs the tube. A student concludes 'air is made by warmth and destroyed by cold.' The accurate description of what the SAME trapped air does is:

- (A) warmth genuinely creates new air that bubbles out, and cold genuinely destroys air so water fills the gap
- (B) warming expands the air so it pushes out as bubbles, and cooling contracts it so the higher outside pressure pushes water up the tube — the same air throughout, neither created nor destroyed
- (C) the water itself boils into bubbles when warm and shrinks back when cooled, while the air does nothing
- (D) the glass flask swells when warm, expelling air, then shrinks when cool, sucking water in

12. A weather map shows a high-pressure centre of 1020 and a low-pressure centre of 1000 sitting 400 km apart. A second map of another region shows a high of 1016 and a low of 1004 sitting 200 km apart. A student must say which region has the stronger wind between its centres. The correct comparison is:

- (A) the first region, because its larger 20-unit pressure difference must drive faster wind regardless of distance
- (B) the second region, because its 12-unit drop over only 200 km is a steeper gradient than the first region's 20-unit drop spread over 400 km
- (C) they are equal, because wind depends only on the low-pressure value and both lows are close to 1000
- (D) neither has wind, because the centres are too far apart in both cases for air to move between them

13. A diver descends in a lake and feels the water press harder the deeper she goes. A student uses this to reason about air and writes: 'Since pressure grows with depth, the air pressure at the bottom of a deep mine should be lower than at the surface.' The correct correction is:

- (A) the student is right: going deeper into the ground means leaving the air behind, so pressure falls
- (B) air pressure does not change with depth at all, so the mine and the surface have identical pressure
- (C) the analogy fails completely, because water pressure and air pressure follow opposite rules
- (D) the analogy actually predicts the opposite of what the student wrote: deeper in the air ocean means more air above you, so air pressure at the bottom of a deep mine is higher than at the surface, just as deeper water presses harder

14. A balanced rod has an open paper bag hanging from each end. A candle warms the air in the left bag, which rises, tipping the rod. The candle is removed and the left bag is allowed to cool back to room temperature, then a block of ice is held under the SAME left bag. Predict the rod's motion:

- (A) the left bag rises again, because cold air is lighter than warm air and floats upward
- (B) the rod returns to level and stays there, because temperature cannot affect a paper bag's balance
- (C) the left bag sinks below level, because cooling its air makes it contract to a denser, heavier state than the surrounding air, so that end is pulled down
- (D) the left bag rises, because ice releases a cold vapour that buoys the bag upward

15. Two long ships steam side by side a short distance apart through still water and are drawn together until they nearly collide. A student says the water between them is squeezed and pushes outward, forcing the ships apart first. Identify the true cause and the flaw:

(A) the water rushing through the narrow gap moves faster and so exerts lower pressure, and the higher outside-water pressure on the far sides pushes the ships together — the flaw is treating fast-moving fluid as a high-pressure pusher

(B) the ships create a vacuum by removing all the water from the gap, and nothing is left to hold them apart

(C) the moving ships magnetise the water, which then attracts the two steel hulls together

(D) the wakes behind the ships pull them backward and only seem to draw them together

16. A coastal town has a reliable afternoon sea breeze. A long total solar eclipse darkens the sky near midday for several minutes, and the land suddenly stops being heated; the sea, holding its warmth, is essentially unchanged for those minutes. During the darkness the sea breeze is expected to:

(A) reverse instantly into a strong land breeze, because the land becomes colder than the sea within those few seconds

(B) strengthen, because sudden darkness always increases wind speed

(C) weaken, because the land stops out-heating the sea, so the pressure difference that normally drives the sea breeze fades

(D) stay exactly the same, because a sea breeze, once started, runs on its own and ignores heating

17. A student reasons: 'In summer the land is hotter than the sea, so the monsoon blows from sea to land. Therefore if some year the sea became hotter than the land in summer, the rain-bearing wind would simply stop.' Evaluate this prediction:

(A) the prediction is exactly right: the wind would stop dead and there would be no breeze in either direction

(B) the rains would actually be heavier, because a hotter sea always sends more moisture onto the land

(C) the prediction is incomplete: if the sea became hotter than the land, the surface wind would not just stop but tend to reverse, blowing from the cooler land toward the lower-pressure warmer sea, and the moist sea-to-land rains would fail

(D) nothing would change, because the monsoon direction is fixed by the calendar and ignores temperature

18. A student lists three ingredients for a thunderstorm: warm surface air that can rise fast, abundant water vapour, and clear height above for tall clouds to build. He is told that exactly one change would most reliably PREVENT the storm. The change with the strongest effect is:

(A) adding even more water vapour to the warm rising air

(B) making the surface air still warmer so it rises faster

(C) replacing the warm, moist surface air with cold, dry air, since that removes both the rising-engine and the vapour at once

(D) clearing away clouds overhead to give still more height for tall clouds to grow

19. Two cyclones X and Y sit over the same warm sea. X's central pressure is 950 and falling; Y's has steadied at 985. A forecaster must judge which threatens a coast more and why. The sound reasoning is:

(A) Y is the greater threat, because its higher central pressure throws the surrounding air outward more forcefully

(B) X is the greater threat, because its much deeper and still-deepening central low creates a bigger pressure difference with its surroundings, pulling in faster eye-wall winds than Y's shallower, steady low

(C) they threaten the coast equally, because two cyclones over the same sea must behave identically

(D) neither threatens the coast, because a cyclone's central pressure is unrelated to its wind speed

20. A student claims a balloon full of air weighs the same as an empty (deflated) one because 'air is invisible and weightless.' A teacher sets up a sensitive balance to settle it. Using the chapter's idea about air, the correct outcome and reason are:

(A) the two balloons weigh exactly the same, confirming that air is weightless

(B) the inflated balloon weighs less, because the air inside makes it buoyant and lifts it

(C) the inflated balloon weighs slightly more, because air has real weight, and that very weight is the reason the atmosphere presses on everything as air pressure

(D) the inflated balloon weighs more only because the stretched rubber is heavier, not because of the air

21. A roof in a cyclone-prone area is at risk because fast wind blowing across its top can lift it off. An engineer proposes drilling small vents through the roof so air can move between inside and outside. Using the fast-air-lower-pressure idea, the vents would:

(A) increase the danger, because any hole lets the wind get under the roof and blow it up like a sail

(B) have no effect, because wind over a roof presses it straight down rather than lifting it

(C) reduce the lifting danger, because letting the inside pressure bleed toward the lowered outside pressure shrinks the up-versus-down pressure difference that tears the roof off

(D) make the roof heavier, so it can no longer be lifted at all

22. A tall glass chimney stands on a table with a candle burning inside near its base; a steady upward draught keeps the flame bright. A student blows a fast horizontal stream of air across the open TOP of the chimney. The flame's behaviour should be:

- (A) it burns even brighter, because the fast air across the top lowers the pressure there, strengthening the upward draught that pulls warm air out and fresh air in to the flame
- (B) it dims, because the fast wind across the top raises the pressure and pushes air down onto the flame
- (C) it is unaffected, because wind across the top of the chimney cannot reach a flame near the base
- (D) it goes out, because blowing across the top blocks the chimney mouth and traps the smoke

23. A ship's log as it sails straight through a large cyclone reads: fierce wind from the north, then a stretch of eerie calm with patches of clear sky overhead, then fierce wind again from the SOUTH. A crew member insists two separate storms passed. The correct single-storm explanation is:

- (A) the leading eye wall gave the first fierce wind, the calm clear eye then passed over the ship, and the trailing eye wall gave fierce wind from the opposite side as the storm continued on its track
- (B) two different cyclones happened to cross the ship one right after the other
- (C) the cyclone paused, rested, and then started up again from the beginning
- (D) the wind simply reversed for no physical reason, as storms often do

24. A region of warm, low-pressure air sits next to a region of cool, high-pressure air. A student must state both the surface wind direction AND what happens to the warm air aloft, then say what would change if the warm region's air were suddenly cooled to match the cool region. The fully correct answer is:

- (A) surface air flows from the warm region to the cool region while the warm air sinks; cooling the warm region would speed the wind up
- (B) no surface air moves and the warm air stays put; cooling the warm region would start a strong wind
- (C) surface air flows toward the warm region while the warm air sinks because it is heavier; cooling the warm region would have no effect
- (D) surface air flows from the cool high-pressure region toward the warm low-pressure region while the warm air rises; if the warm region cooled to match, the pressure difference would vanish and the surface wind would die away

25. A coastal town that once lost thousands of lives to a cyclone now loses very few in storms of similar strength. A reporter lists four changes. Which is the chapter's MAIN reason for the drop, and why are the others not the main cause?

- (A) the cyclones striking the coast have grown gentler over the years, so they simply do less harm
- (B) the sea has stopped surging over the low coast during these storms

- (C) almost everyone has permanently moved away from the coastal district
- (D) weather satellites and radar now give accurate warnings a day or two ahead, so people reach shelters in time — the cyclones themselves are not gentler, the sea still surges, and people still live there

26. A student measures the time between lightning and thunder at three moments in a passing storm: first 9 seconds, then 6 seconds, then 9 seconds again. Interpreting the down-then-up pattern, the storm most likely:

- (A) moved steadily closer the whole time, because any change in the gap means approach
- (B) approached for a while and then began moving away again, since the gap first shrank (closer) and then grew (farther) as the sound's travel distance changed
- (C) stayed in exactly one spot, because the gap depends only on how bright the lightning is
- (D) had its thunder speed up and then slow down, which alone changed the gaps with no movement

27. A sealed plastic bottle of warm air is capped indoors, then carried up a tall mountain where the outside air is much thinner, and finally left to cool to the cold mountain air temperature. Considering BOTH the altitude change and the cooling, the bottle's most likely state is:

- (A) swollen outward, because thin mountain air lets the inside air expand freely
- (B) unchanged, because the thinner outside air and the cooling exactly cancel and leave it as it was
- (C) noticeably dented inward, because both effects work the same way — the thinner mountain air presses in less, yet the cooling contracts and lowers the inside air's pressure, so the inside loses to the outside push
- (D) swollen outward, because cooling makes trapped air expand to fill more space

28. On the open sea a cyclone drives a great mound of water onto a low coast that floods far worse than the highest ordinary tide. A student asks what makes the storm surge so much worse than a normal high tide. The best answer is:

- (A) the cyclone's very low central pressure lets the sea rise and its fierce winds heap up and push a huge mass of sea water far inland, well beyond what a normal tide reaches
- (B) the surge is only slow rainwater that gradually collects on the low coast over many hours
- (C) during a surge the sea level actually drops, exposing dry land that floods later
- (D) the surge is fresh river water flowing backward from upstream into the sea

29. A demonstration shows a slanted lamp on a globe so the equator gets near head-on light and the poles only grazing light. A student must connect this to Earth's great wind belts. The correct chain is:

(A) the slant makes the poles warmer than the equator, so winds blow only from the equator outward to the poles

(B) the slant warms every part of the globe equally, so no wind belts can form anywhere

(C) the slant makes the equator far warmer than the poles, so warm equatorial air rises while cold polar air sinks, setting up giant circulating winds that carry heat around the planet

(D) the slant makes equatorial air lose its weight, and that weightlessness is what starts the wind belts

30. A student arranges a growing cyclone's self-feeding loop but scrambles the steps: (i) faster wind spirals in toward the centre; (ii) condensing vapour releases heat that deepens the central low; (iii) the Sun warms the sea so warm moist air rises; (iv) the deeper low and rising air pull still more moist air upward. The order in which each builds on the previous is:

(A) (iii) first, then (ii), then (i), then (iv) (B) (i) first, then (ii), then (iii), then (iv)

(C) (ii) first, then (iii), then (iv), then (i) (D) (iv) first, then (i), then (ii), then (iii)

31. A teacher fills a tumbler to the brim, covers it with a stiff card, and inverts it; the water stays. She asks: if the same setup is done but a small bubble of air is deliberately trapped above the water before inverting, the card is now more likely to fall because:

(A) the trapped air sits on top of the water (which is now at the bottom) and its pressure adds to the water's weight pressing down on the card, so the upward air pressure may no longer be enough to hold it

(B) the smaller amount of water left is somehow heavier than a completely full tumbler of water

(C) air pressure only acts to hold up completely full containers and ignores any with a bubble

(D) the trapped bubble glues itself to the card and physically drags it downward

32. An anemometer, a barometer and a thermometer share one weather station. Over an afternoon the anemometer cups spin faster and faster while the barometer falls steadily; the thermometer barely changes. Reading all three together, the safest conclusion is:

(A) calm, settled fair weather is on the way, since the instruments are all behaving routinely

(B) the air is simply heating up, which is what makes the anemometer cups spin faster

(C) rain is falling right now, which is the only thing a barometer can ever indicate

(D) a storm or low-pressure system is likely approaching, since falling pressure together with strengthening wind is the classic signature, and the steady temperature rules out a mere heat change

33. A student holds a paper strip below her lip and blows hard across its TOP; it lifts. She then flips the strip and blows hard across its BOTTOM surface while the top stays still. The free end now:

- (A) lifts up just as before, because blowing on paper always raises it whichever side you choose
- (B) stays perfectly still, because which surface you blow over makes no difference
- (C) lifts even more strongly, because blowing underneath gives the strip extra lift
- (D) is pushed downward, because the fast air is now beneath the strip at lower pressure while the still higher-pressure air above presses it down

34. A student writes: 'Wind blows from cold places to hot places because cold air is attracted to warmth.' A teacher says the direction is often right but the reason is wrong. The teacher's precise correction is:

- (A) wind blows from high pressure to low pressure; hot regions usually become low-pressure as their air rises, so cooler higher-pressure air does flow toward them — but pressure difference, not any attraction to heat, is the real driver
- (B) the student is fully right, because cold air really is attracted to warmth like iron to a magnet
- (C) wind always blows from hot to cold, so both the stated direction and the reason are wrong
- (D) wind direction has nothing to do with temperature or with pressure at all

35. Inside a developing thunderstorm cloud, warm moist air surges upward while water drops and ice fall back through it. A student asks which single feature is the chief ENGINE of the storm's violence. The best answer is:

- (A) cold, dry air quietly sinking to the ground with no moisture taking part
- (B) lightning forming first and then dragging the air upward behind it
- (C) the warm ground touching and heating the cloud directly from below
- (D) the rapidly rising warm, moist air together with the water and ice falling back through it, which drives the cloud's churning and energy

36. A teacher boils water in a metal can to fill it with steam, seals it, and cools it; it crumples. She asks what would happen if, just before cooling, she punched a wide hole in the can and left it open during cooling. The can would:

- (A) not crumple, because outside air would flow in through the hole as the steam cooled, keeping the inside and outside pressures equal so no crushing difference could build
- (B) crumple even harder, because the open hole lets the steam rush out faster and pull the walls in
- (C) crumple only on the side bearing the hole, while the rest stays straight
- (D) melt around the edges of the hole instead of crumpling anywhere

37. A coastal town installs a huge field of mirrors that, on a test day, reflect away much of the sunlight before it can heat the land; the sea is unaffected. Compared with a normal sunny day, the afternoon sea breeze on the test day is most likely to be:

- (A) stronger, because the mirrors add their own reflected wind to the breeze
- (B) unchanged, because a sea breeze does not depend on how much the land is heated
- (C) reversed into a strong land breeze, because the mirrors make the land hotter than the sea
- (D) weaker, because with the land heated far less the land-versus-sea pressure difference that drives the sea breeze is reduced

38. A student must explain BOTH the direction and the dryness of India's winter winds, which blow from land to sea and carry little rain. The correct combined reason is:

- (A) in winter the sea cools faster than the land, so wet air rushes from the sea onto the dry land
- (B) winter winds blow straight up, so they bring snow rather than rain
- (C) winter winds are wetter than summer winds, because cold air always carries more water than warm air
- (D) in winter the land cools more than the sea, so higher pressure over the cool land pushes air out toward the sea, and that air, coming off dry land rather than picking up sea moisture, is dry

39. A student claims a cyclone's calm, clear eye must be the most dangerous part because it sits at the very centre 'where everything is strongest.' Tested against the real structure, the correct general lesson is:

- (A) being at the geometric centre does not make a place the most violent; in a cyclone the violence peaks in the eye-wall ring around the calm eye, so the eye itself is the least dangerous part of the inner storm
- (B) the eye really is the most violent part, because the centre of any storm always has the strongest winds
- (C) the calm eye and the violent eye wall are equally dangerous, since both lie near the centre
- (D) the outermost cloud bands are the most violent, while the eye wall is the calmest part

40. A student is told that a strong cyclone, a hurricane, and a typhoon are 'three different storms of three fixed sizes, the typhoon always largest.' Pinpoint the error:

- (A) the only error is the order: the hurricane, not the typhoon, is always the largest of the three
- (B) there is no error at all; the three are genuinely different storms of fixed increasing size
- (C) the only error is that a typhoon is actually a kind of tornado, unrelated to the other two
- (D) they are not three different storms or three fixed sizes — cyclone, hurricane and typhoon are simply regional names for the same kind of tropical storm, so none is fixed as the largest

41. A patch of atmosphere above a sun-baked plain is heated strongly from below for several hours, while a neighbouring shaded patch stays cool. Tracking the surface pressure under each patch over the afternoon, you would expect:

- (A) the heated patch's pressure to rise, because the warmed air becomes heavier and presses down harder
- (B) the pressure under the heated patch to fall, because its warmed air expands and rises away leaving less air pressing down, while the cool shaded patch keeps higher surface pressure
- (C) both patches' pressure to fall equally, because heating one patch lowers pressure everywhere alike
- (D) the heated patch's pressure to drop to zero, because heating removes all the air above the plain

42. Two identical sealed jars of air are prepared in a valley. One is carried to a high mountain peak; the other stays in the valley. The trapped air in the peak jar is the same amount as before. Comparing the NET strain on the two jars' walls:

- (A) the peak jar feels a net inward squeeze, because mountains have stronger air pressure than valleys
- (B) both jars feel exactly the same strain, because sealed air never changes no matter where it goes
- (C) the peak jar feels a net outward strain, because the outside air pushing in is now thinner and weaker while the inside air still pushes out as strongly as before
- (D) the peak jar feels no strain at all, because air loses all its pressure once you climb a mountain

43. A cyclone warning is broadcast. A family debates four actions. Using cyclone-safety reasoning, which single action is BOTH safe and correctly justified?

- (A) store clean drinking water in advance, because the storm may contaminate or cut off the normal water supply
- (B) throw open all doors and windows so the strong wind can blow straight through the house
- (C) go down to the seashore to watch and photograph the giant waves up close
- (D) shelter under the tallest lone tree in the open, because tall trees block the wind best

44. A student must label, from the calm centre of a cyclone outward, three zones by their wind and pressure. The correct centre-to-edge labelling is:

- (A) a violent, high-pressure eye, then a calm eye wall, then a still outer ring with no clouds
- (B) a calm, low-pressure eye, then a violent eye wall of the fastest winds and heaviest rain, then outer bands of spiralling clouds

(C) a calm high-pressure eye, then calm clouds, then a violent outer rim that is the fiercest part

(D) a calm eye, then a calm eye wall, then a calm edge, with no violent zone anywhere

45. A student lists three things a cyclone needs to keep intensifying over the ocean: warm sea water, plenty of water vapour rising and condensing, and a low-pressure centre for wind to rush into. He must name the single change that would most directly STOP the intensification. That change is:

(A) increasing the amount of water vapour available to rise and condense

(B) deepening the low pressure at the centre still further

(C) moving the storm over much cooler water, so little warm moist air rises and little condensation heat is released to deepen the low

(D) supplying still more warm moist air to feed the rising column

46. A teacher seals a balloon over a flask's mouth, dips the flask in hot water (the balloon swells), then moves it straight into ice water. A student must predict the balloon AND describe what the inside air's pressure does relative to the outside. The correct full answer is:

(A) the balloon swells even further, because cold air expands to take up more room

(B) the balloon stays as swollen as in the hot water, because cooling has no effect once the air was warmed

(C) the balloon collapses, but only because the cold makes the rubber stretch and pull itself inward

(D) the balloon collapses inward, because the cooling air contracts and its pressure falls below the outside air pressure, so the outside air pushes the balloon in

47. A coastal town gets a steady afternoon sea breeze on clear days. On one heavily overcast afternoon the clouds keep the land from heating much; on another afternoon a cold spell leaves the land actually colder than the sea. On these two days the afternoon sea breeze should respectively be:

(A) stronger on both days, because cloud cover and cold both increase wind speed

(B) exactly the same on both days, because the sea breeze never depends on the land's temperature

(C) a stronger sea breeze on the cold day, because cold land air rushes out toward the warm sea as a sea breeze

(D) weaker on the overcast day and reversed (a land breeze) on the cold-spell day, because the breeze depends on the land being warmer than the sea, which is reduced on one day and flipped on the other

48. A meteorologist watches the barometer at a port fall unusually fast through the morning while fishermen radio that the wind far out at sea is rising. Putting the two together, the safest advisory is to:

- (A) announce that fine, settled weather is guaranteed for the rest of the day
- (B) warn that a storm or cyclone may be approaching and advise boats to return, because falling pressure together with rising wind is the signature of a low-pressure system
- (C) tell the fishermen the changes are meaningless and to keep fishing far out
- (D) predict that the temperature is about to drop below freezing at the port

49. A light ball hovers in the upward stream of a vertical hair dryer and springs back when nudged sideways. The dryer is then tilted to point its stream at about 45 degrees. The ball tends to keep hovering WITHIN the slanted stream because:

- (A) the ball is weightless and simply floats wherever the dryer happens to point
- (B) the air in the stream is sticky and glues the ball onto the dryer's path
- (C) tilting the dryer adds a magnetic pull that keeps the ball lined up with the stream
- (D) wherever the fast stream goes, its pressure is lower, so the slower higher-pressure air around it pushes the ball back into the stream, now along the slanted direction

50. A student watches a thunderstorm and jots four events out of order: (i) drops grow heavy and fall as rain; (ii) warm moist air rises and cools; (iii) water vapour condenses into towering tall clouds; (iv) the ground and surface air are warmed on a hot humid day. The order in which they actually occur is:

- (A) (iv) first, then (ii), then (iii), then (i) (B) (i) first, then (ii), then (iii), then (iv)
- (C) (iii) first, then (i), then (iv), then (ii) (D) (ii) first, then (iv), then (i), then (iii)

51. A weather buff records four facts about a storm: a roughly circular shape only a few hundred metres across, an extremely fast whirling column reaching down from a cloud to the ground, a very short life of minutes, and great destruction along a narrow track. This storm is best identified as a:

- (A) full tropical cyclone hundreds of kilometres across
- (B) tornado
- (C) gentle sea breeze blowing in from the coast
- (D) wide, slow-moving high-pressure region

52. A student must explain why a sea breeze typically blows during the DAY and a land breeze at NIGHT, and what single fact underlies both. The deepest correct answer is:

- (A) the Sun directly blows the wind toward shore by day and away from shore by night
- (B) the sea changes temperature faster than the land, which sets both breeze directions
- (C) land changes temperature faster than the sea, so by day the quickly-heated land draws sea air in, and by night the quickly-cooled land lets the air flow back out to sea
- (D) winds simply reverse at sunrise and sunset on a fixed schedule with no link to heating

53. A student is asked why the deep, very low-pressure centre of a cyclone matters so much for its wind strength. The clearest cause-and-effect answer is:

- (A) the deeper the central low, the bigger the pressure difference between the centre and its surroundings, so the faster the surrounding air rushes inward as wind
- (B) a deep central low makes the surrounding air go still, so the wind dies down near the centre
- (C) a deep central low pushes the surrounding air outward, which is what we feel as wind
- (D) the centre's pressure has no link to the wind speed, which is set only by the sea's temperature

54. A coastal builder is warned that fast wind across a roof can tear it off. He compares three designs. Using the fast-air-lower-pressure idea, the design that BEST resists lifting is:

- (A) a very light, loosely laid roof, because light roofs are too small for the wind to lift
- (B) a heavy, well-anchored, low-profile roof, because firmly fixing a low roof down counters the upward lift created when fast wind lowers the pressure above it
- (C) a tall, broad roof left unanchored, because its sheer size alone holds it down
- (D) any roof at all, because fast wind over a roof actually presses it straight down, not up

55. A student lights a candle just inside the lower opening of a tall glass chimney standing on a table; the flame burns brightly with a steady upward draught. A card is slipped over the TOP opening, sealing it. The flame would most likely:

- (A) burn brighter, because the trapped warm air now feeds the flame more strongly
- (B) dim or go out, because sealing the top stops the warm air rising out and fresh air drawing in, cutting off the draught the flame needs
- (C) be unaffected, because a flame does not depend on air moving through the chimney
- (D) turn into faster-rising smoke, because blocking the top speeds the draught

56. A teacher fills a tumbler to the brim, covers it with a light card, inverts it, and the water stays. She asks: if she instead used a small, very heavy flat metal plate of the same size as the cover, what would most likely happen and why?

- (A) the plate is held even more firmly, because heavier covers are gripped harder by air pressure
- (B) nothing changes, because the cover's own weight never matters to this experiment
- (C) the plate floats away upward, because heavy metal is lifted by air pressure
- (D) the plate may fall, because its extra weight adds to the downward pull, and the total downward force may now exceed the upward push of air pressure that easily held the light card

57. A cyclone that has been intensifying over the open ocean crosses the coastline and within a day weakens sharply. Among four offered reasons, the chapter's correct one is:

- (A) over land the storm becomes colder than outer space and freezes solid
- (B) the buildings on land act as a solid wall that physically blocks all the wind

(C) over land the storm is cut off from the warm, moist sea air that supplied the heat and vapour feeding it, so its self-feeding cycle is starved

(D) the eye instantly snaps shut the moment the storm touches soil, stopping the wind

58. Two cyclones, M and N, form over the same warm sea on the same day. M's eye-wall winds are recorded at about 200 km/h and N's at about 120 km/h. Using the chapter's link between central pressure and wind, the most reasonable inference is:

(A) N most likely has the lower central pressure, because slower winds always go with a deeper low

(B) both must have exactly the same central pressure, because they formed over the same sea on the same day

(C) M most likely has the higher central pressure, because higher pressure flings the eye-wall winds outward faster

(D) M most likely has the lower (deeper) central pressure, because a deeper central low creates a bigger pressure difference that drives the faster eye-wall winds

59. A student is asked the single deepest fact that drives BOTH the moist sea-to-land summer monsoon and the dry land-to-sea winter wind over India. The most fundamental correct answer is:

(A) land changes temperature faster than the sea, so in summer the hotter land draws sea air in and in winter the cooler land pushes air out to sea

(B) the Moon's gravity reverses the prevailing winds between summer and winter

(C) the sea changes temperature faster than the land, and that sets both wind directions

(D) the monsoon winds simply reverse on a fixed calendar date with no link to any heating

60. A student must order four whirling-air facts to match a cyclone passing directly over a coastal town: (i) a sudden eerie calm with a glimpse of clear sky; (ii) fierce wind from one direction; (iii) fierce wind from the opposite direction; (iv) the wind and rain finally easing as the outer bands move off. From the town's first experience to its last, the order is:

(A) (i) first, then (ii), then (iii), then (iv) (B) (ii) first, then (iii), then (i), then (iv)

(C) (iv) first, then (ii), then (i), then (iii) (D) (ii) first, then (i), then (iii), then (iv)

Solutions — Science (Class 7), Chapter 8: Winds, Storms and Cyclones — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	C	31	A	41	B	51	B
2	B	12	B	22	A	32	D	42	C	52	C
3	C	13	D	23	A	33	D	43	A	53	A
4	B	14	C	24	D	34	A	44	B	54	B
5	A	15	A	25	D	35	D	45	C	55	B
6	C	16	C	26	B	36	A	46	D	56	D
7	B	17	C	27	C	37	D	47	D	57	C
8	A	18	C	28	A	38	D	48	B	58	D
9	B	19	B	29	C	39	A	49	D	59	A
10	C	20	C	30	A	40	D	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) The globe is sealed and rigid, so the quantity of air inside is fixed and its total weight cannot change — the balance reads the same. The popular phrase 'warm air is lighter' refers to density: heating makes the air expand, so the same amount of air occupies more space and each cubic-metre weighs less, which is why a warmed parcel floats up through cooler surroundings. It does not mean heat erases weight (the first wrong choice), nor that heat is a kind of mass that adds to the reading (the second), nor that air trapped inside a closed globe can push its own container's weight off a balance (the third). The lesson is that 'lighter' here means lower density, not less total weight.

2. (B) When you fill a fixed volume with cold air, the air is denser, so more air fits into that volume than when you fill it warm. Balloon P was filled cold, so it holds more air than Q which was filled warm to the same size. After both reach room temperature, P — having more air sealed in — is the more tightly filled and heavier one. The choice claiming Q holds more confuses expansion with quantity (warm air takes more space, so you need less of it to fill a given size, not more). 'Same size means same amount' ignores that density depends

on temperature at filling. And cold air does not leave empty gaps; it simply packs more densely. So denser cold filling means more air, making P the answer.

3. (C) Cooling plain air only makes it contract a little, so its pressure drops only slightly and the outside air barely wins — a small dent. But steam, on cooling, condenses into a minute volume of liquid water, removing almost all the gas inside; the inside pressure plummets, and the much higher outside air pressure crushes the can. The crush is entirely about the inside-versus-outside pressure imbalance. The student's 'steam is heavier' idea is the flaw: weight of the gas inside is irrelevant; condensation, not weight, empties the can. Boiling does not weaken the metal (the can crushes because of the pressure gap, not soft walls), and ordinary air can certainly be cooled by ice water — it just doesn't vanish the way condensing steam does.

4. (B) Wind speed depends on the pressure gradient — how big a drop falls over how short a distance — not on which station is highest or on the total end-to-end difference. Taking a full-width gap as one unit: first-to-second drops 6 over a full gap (6 per unit); second-to-third drops 4 over a full gap (4 per unit); third-to-fourth drops 6 over a half-width gap, which is 12 per unit — by far the steepest. So the fastest wind is between the third and fourth stations. The 'highest pressure makes fastest wind' choice confuses pressure value with gradient. 'Closest to the centre' has nothing to do with speed. And the end-to-end first-to-fourth difference, though largest in total, is spread over the longest distance, giving a gentle gradient. The steep drop over the narrow third-to-fourth gap wins.

5. (A) Land heats faster than water, so by noon the air over the field is warmer. That warm air expands, drops in density, and rises; as it leaves, less air presses on the field's surface, so the surface pressure there falls. Wind blows from higher pressure (over the cooler pond) toward lower pressure (over the field), so the surface breeze moves from pond to field. The choice that says water heats faster reverses the well-known land-versus-water rule. The 'warm air is heavier and sinks' choice is the classic density error — warm air is less dense and rises. And the two surfaces do not heat equally; land's faster heating is exactly what sets up the pressure difference that drives the breeze.

6. (C) Where the card narrows the gap, the air is forced to move faster, and faster-moving air has lower pressure. With lower pressure on the card side and ordinary higher still-air pressure on the open side, the net push moves the ball toward the card — the same Bernoulli effect behind balls being drawn together by a stream. The choice that says the card raises pressure on its side reverses the rule (faster air means lower, not higher, pressure). Air does not 'stick and drag' the ball toward the nearest object — that is a vague non-mechanism. And a card beside the ball adds no weight to the ball. The pressure difference set up by the faster channelled air is what moves it.

7. (B) The card is held up because the outside air pressure pushing up on it exceeds the weight of the water pushing down. On a mountain peak there is less air above, so the

outside air pressure is lower and pushes up less. The water's weight is the same as before, so the margin holding the card shrinks and the card is more likely to fall. 'Thin air clings tighter' invents a non-existent stickiness. 'Pressure is the same everywhere' is simply false — it falls with altitude. 'Water cannot fall upside down' ignores that only the air pressure holds it; remove enough of that push and it falls. So lower peak pressure makes the card more likely to drop.

8. (A) A cyclone feeds on warm moist air rising from a warm sea; condensation of that vapour releases heat that deepens the central low and pulls in faster wind. The half over land loses that supply of warm moist air, so part of the engine is starved and the storm weakens overall — this is why cyclones fade after landfall. Land does not act as a surface to 'spin against' for extra speed. A cyclone is not self-sufficient; cut off its moisture and it dies, so it certainly still needs the sea. And partial overlap with land does not cleanly split it into two equal storms. Starving half the moisture supply means net weakening.

9. (B) Reading the sizes: full cyclone 500 km, eye 30 km, thunderstorm cell 10 km, tornado 0.3 km. From largest to smallest that is cyclone (500), eye (30), cell (10), tornado (0.3). The choice that puts the tornado first commits the classic error of ranking the narrow, violent tornado as the biggest — it is in fact the smallest. The third option swaps the eye and the cell (30 km vs 10 km, so the eye is bigger). The last option puts the eye ahead of the whole cyclone, which is impossible since the eye is just the small core of the cyclone. Careful reading of the figures gives cyclone, eye, cell, tornado.

10. (C) Light reaches us almost instantly, so the delay before the thunder is essentially the time the sound takes to travel from the storm to the listener. A steady gap means the distance held constant; a sudden drop to a shorter gap means the sound now has less distance to cover — the storm has come closer. The 'moved away' choice reverses the relationship (farther storms give longer, not shorter, gaps). Brightness of the lightning does not change the travel time of the sound. And the speed of sound does not jump in the middle of a storm; the change in the gap reflects the changing distance, so the storm moved closer.

11. (B) The trapped air is a fixed amount throughout. Warming makes it expand, so some pushes out of the tube as bubbles; cooling makes it contract, lowering its pressure so the higher outside pressure drives water up the tube to take the freed space. No air is created or destroyed — the student's conclusion is the misconception the demo is designed to defeat. The water does not boil at hand-warming temperatures (and the bubbles are escaping air, not steam). The glass flask's tiny expansion is negligible and is not what moves the water; it is the air's expansion and contraction that does. So the same air simply expands and contracts.

12. (B) Wind speed tracks the pressure gradient: pressure difference divided by distance. The first region drops 20 units over 400 km, about 0.05 units per km. The second drops 12 units over 200 km, about 0.06 units per km — steeper. So the second region, despite a

smaller total difference, has the stronger gradient and the faster wind. The choice favouring the first region looks only at the total difference and ignores the distance over which it falls. Wind does not depend on the low value alone; it depends on the difference and the spacing. And distances of a few hundred km are entirely normal for wind to blow across, so 'no wind' is wrong.

13. (D) We live at the bottom of an ocean of air, and pressure rises as you go deeper into that ocean because more air lies above you — exactly the diver's experience that deeper water presses harder. Descending into a deep mine puts even more air column above you, so the air pressure there is higher than at the surface, not lower. The student stated the conclusion backwards. Air pressure certainly does change with position in the air column (it falls on mountains, rises in deep places). And the analogy does hold: both fluids press harder the deeper you go, so they follow the same rule, not opposite ones.

14. (C) The candle warmed the left bag's air, expanding it to a less dense, lighter state, so it rose. The ice does the reverse: it cools the air in the left bag, which contracts to a denser, heavier state than its surroundings, so that end is pulled down below level. The first wrong choice repeats the density error — cold air is denser and heavier, not lighter. Temperature certainly affects the balance, as the candle already showed, so 'returns to level and stays' is wrong. And ice does not emit a buoyant 'cold vapour' that lifts things. Cooling makes the air denser and the bag heavier, so it sinks.

15. (A) Water forced through the narrow gap between the hulls speeds up, and faster-moving fluid has lower pressure. The higher pressure of the slower water on the ships' outer sides then pushes them inward toward each other — the same principle as the two balls drawn together by a stream of air. The student's flaw is exactly the Bernoulli reversal: fast fluid is at lower, not higher, pressure, so it does not push outward. The ships do not remove 'all' the water to make a vacuum; water still flows, just faster and at lower pressure. There is no magnetism involved. And the drawing-together is a sideways pressure effect, not a backward wake pull. Lower pressure in the fast gap is the cause.

16. (C) The sea breeze blows because the strongly heated land develops lower pressure than the cooler sea, drawing sea air inland. During the eclipse the land stops being heated, so it stops out-heating the sea; the pressure difference shrinks and the breeze weakens. It does not reverse within seconds — the land cannot drop below the sea's temperature in a few minutes; the land merely stops getting hotter, it does not instantly become colder than the warm sea. Darkness itself does not increase wind. And the breeze is not self-sustaining; it depends moment-to-moment on the temperature difference, so removing the heating weakens it.

17. (C) The monsoon's sea-to-land summer wind depends on the land being hotter and therefore lower in pressure than the sea, drawing sea air in. If instead the sea became hotter than the land, the lower pressure would form over the sea; surface air would tend to flow from the cooler higher-pressure land toward the warmer lower-pressure sea — a

reversal — and the moist sea-to-land rains would fail. So the wind doesn't merely stop; the driving difference flips, so the tendency is to reverse. A hotter sea pushing 'more moisture onto land' ignores that the wind would now blow the other way. And the monsoon is driven by the land-sea temperature difference, not fixed by the calendar regardless of temperature.

18. (C) A thunderstorm needs warm air that rises rapidly and plenty of moisture to condense into towering clouds. Swapping in cold, dry air defeats two requirements together: cold air does not rise quickly, and dry air supplies little vapour to condense — so the storm is starved most reliably. Adding more vapour or making the air warmer would only feed the storm more, not prevent it. Clearing more height overhead also helps the storm grow, not stop. Removing the warm moist rising air is the one change that disables the storm's engine and fuel at once.

19. (B) The deeper the central low, the larger the pressure difference between the calm centre and the surroundings, and the faster the surrounding air rushes inward as wind. X at 950 and still falling has a deeper, deepening low than Y steady at 985, so X pulls in faster eye-wall winds and is the greater threat. The Y choice reverses the physics — a higher central pressure means a weaker inward pull, and cyclones draw air inward, not throw it outward. Two cyclones over the same sea need not be identical; their central pressures differ. And central pressure is tightly linked to wind speed, so 'neither is dangerous' is wrong. X's deeper low wins.

20. (C) Air has weight; a balloon full of air contains more air than a deflated one, so on a sensitive balance the inflated balloon reads slightly heavier. This same weight of the air column above us is what produces atmospheric pressure. The 'weightless' conclusion is the misconception the experiment refutes. The 'weighs less because buoyant' choice confuses density-driven floating with total weight — the air inside still has mass and adds to the reading. And the extra weight is the air itself, not merely the rubber, which is the same balloon in both states. So the inflated balloon is heavier because air has weight.

21. (C) Fast wind across the top lowers the pressure above the roof, while the still air inside the house remains at higher pressure, so the roof is pushed up — the lifting danger. Small vents let the trapped inside air bleed toward the lowered outside pressure, shrinking the inside-outside pressure difference and so reducing the upward lift. The 'blow it up like a sail' choice imagines the wind getting under and pushing up, but the small vents equalise pressure rather than catch wind. Wind over a roof lowers pressure and lifts it (it does not press it down). And vents do not add weight. Bleeding the pressure difference is what helps.

22. (A) Fast air across the chimney's open top lowers the pressure there. With lower pressure above and ordinary pressure below, the upward draught through the chimney strengthens — warm air is drawn out faster and fresh air is pulled in at the base, feeding the flame, so it burns brighter. The 'raises pressure and pushes down' choice reverses

Bernoulli; fast air lowers pressure. The effect does reach the flame: it works precisely by strengthening the draught through the whole chimney. And blowing across the top does not block it — air streaming over an opening is not a cap. The lowered pressure at the top boosts the draught and the flame.

23. (A) A cyclone has a calm, clear low-pressure eye surrounded by the violent eye wall. Passing through the centre, a ship meets the leading eye wall (fierce wind), then the calm clear eye with breaks of sky, then the trailing eye wall, whose winds blow from the opposite direction because the storm's circulation is on the other side of the centre now. This is one storm, not two passing in succession. Cyclones do not pause and restart. And the reversal is not random — it is exactly what the eye-and-eye-wall structure predicts. One cyclone with a calm eye between two eye walls explains the whole log.

24. (D) Wind blows from high to low pressure, so surface air moves from the cool high-pressure side to the warm low-pressure side, while the warm air, being less dense, rises. If the warm region were cooled to match the cool one, both would be at the same pressure, the difference driving the wind would disappear, and the surface wind would die. The first wrong choice reverses the wind direction and has warm air sinking. The 'no air moves' choice ignores the existing pressure difference. And warm air does not sink because it is 'heavier' — it is less dense and rises; removing the temperature difference is exactly what stops the wind, so 'no effect' is wrong.

25. (D) The decisive modern change is early warning: satellites and radar detect and track cyclones a day or two ahead, letting authorities evacuate people to shelters before landfall. Cyclones of similar strength are not inherently gentler now — the storms are as strong; what changed is the warning and response. Storm surges still occur during cyclones; they have not stopped. And coastal districts remain populated, so mass permanent abandonment is not the reason. Accurate advance warning saving lives is the main cause.

26. (B) The gap measures how long the sound takes to travel from the storm, so it tracks distance. The gap shrinking from 9 to 6 seconds means the storm came closer; the gap then growing back to 9 means it moved away again — approach then retreat. 'Steadily closer the whole time' ignores that the gap grew again at the end. The gap depends on distance, not on the lightning's brightness, so 'stayed in one spot' is wrong. And the speed of sound does not speed up and slow down mid-storm; the changing gap reflects the storm's changing distance. So it approached, then receded.

27. (C) Two things happen. The thin mountain air outside presses in less than indoor air did, which by itself would let the bottle swell. But the cooling contracts the trapped air and lowers its inside pressure sharply. The cooling effect dominates here: the inside pressure drops below even the reduced outside pressure, so the outside air still wins and the bottle dents inward. The 'swollen because thin air lets it expand' choice forgets the cooling that shrinks the inside air. The two effects do not exactly cancel — the strong cooling outweighs

the modest pressure drop outside. And cooling contracts trapped air, it does not expand it. So the bottle dents inward.

28. (A) A storm surge is sea water heaped up by the cyclone: the very low central pressure lets the sea surface rise, and the fierce winds pile up and drive that water onto and far across the low coast — much farther and higher than an ordinary tide. It is not slow rainwater collecting (it is a sudden wind-and-pressure-driven mound of sea water). The sea level rises in a surge, it does not drop. And it is sea water driven inland, not fresh river water flowing backward. Low pressure plus fierce wind heaping up the sea is what makes the surge so destructive.

29. (C) Head-on light concentrates more heat at the equator; grazing light spreads thin heat at the poles, so the equator is far warmer. Warm equatorial air rises and cold polar air sinks, and air flows between them, setting up the planet-scale wind circulation that redistributes heat. The 'poles warmer' choice reverses the heating. The slant does not warm the globe equally — that uneven heating is the whole point. And the wind belts arise from rising warm and sinking cold air, not from air 'losing its weight.' The equator-warmer-than-poles chain is correct.

30. (A) The loop starts with the Sun warming the sea so warm moist air rises (iii). As that vapour condenses higher up, it releases heat that deepens the central low (ii). The deepened low pulls surrounding air inward faster, spiralling toward the centre (i). That deeper low and the rising air then draw still more moist air upward (iv), feeding the cycle onward. So the build-up is iii, ii, i, iv. The other orderings start mid-cycle — with wind already spiralling, or with condensation before any rising air, or with the feedback step before the initial warming — none of which can be the starting cause. The Sun-warmed rising air must come first.

31. (A) With the tumbler inverted, the water is at the bottom against the card and the trapped air bubble is above it. That air pushes down on the water with its own pressure, adding to the water's weight on the card. The upward air pressure on the card is unchanged, so the total downward push can now exceed it and the card falls. The 'less water is heavier' choice is self-contradictory. Air pressure does not magically only work on completely full containers — it acts the same, but the added downward push from the bubble tips the balance. And the bubble does not 'glue and drag' the card; it presses down through its pressure. The added downward pressure of the trapped air is the reason.

32. (D) The anemometer measures wind speed (faster spin = stronger wind) and the barometer measures pressure (falling = dropping pressure). Strengthening wind plus falling pressure is the textbook approach of a low-pressure system or storm; the near-steady thermometer shows this is not just a temperature change. 'Settled fair weather' contradicts falling pressure with rising wind. Heating does not spin the anemometer — wind does, and the thermometer is steady anyway. And a barometer measures pressure, not present rainfall. Falling pressure with rising wind signals an approaching storm.

33. (D) Fast-moving air has lower pressure. When she blows across the top, the lower pressure is above and the higher still-air pressure below pushes the strip up. When she blows across the bottom, the lower pressure is now below and the higher still-air pressure above presses the strip down. So flipping which side gets the fast air flips the direction of the push. 'Blowing always raises paper' ignores that lift came from low pressure being on the upper side. The side certainly matters — it determines where the low pressure sits. And blowing underneath gives a downward push, not extra lift. So the free end is pushed down.

34. (A) The true driver of wind is the pressure difference: air moves from higher to lower pressure. Hot regions tend to develop low pressure because their warmed air expands and rises, so cooler air at higher pressure does flow toward them — which is why the direction the student named is often correct. But there is no 'attraction to heat'; pressure does the pushing. Air is not magnetically drawn to warmth. The direction is not always hot-to-cold — it depends on where the low pressure forms, which is usually over the hot region. And wind is closely tied to temperature through pressure, so 'nothing to do with either' is wrong. Pressure, set up by heating, is the real cause.

35. (D) A thunderstorm's violence comes from the vigorous up-and-down churning: warm moist air rushing upward and water drops and ice falling back down through it, releasing energy as vapour condenses. Quiet, dry sinking air with no moisture is the opposite of what powers a storm. Lightning is a result of the charge separation produced by this churning, not a first cause that drags the air up. And the ground heats the surface air from below to start the rising, but the cloud's engine is the rising moist air and falling precipitation inside it, not direct contact heating of the cloud. The rising warm moist air with falling water and ice is the engine.

36. (A) The crush comes from a pressure difference: when the steam condenses, the sealed can's inside pressure falls far below the outside, and the outside air crushes it. With a wide hole left open, outside air simply flows in to replace the condensing steam, so the inside pressure stays equal to the outside and no crushing difference forms — the can keeps its shape. Letting steam 'rush out faster' does not pull walls in; the open hole equalises pressure rather than creating a vacuum. The crush, when it happens, is all-around from uniform outside pressure, not one-sided at a hole. And cooling does not melt metal. An open hole equalises the pressure, so no crush.

37. (D) The sea breeze depends on the land heating more than the sea, forming lower pressure over land that draws in sea air. Mirrors that reflect away sunlight keep the land from heating much, so the land-sea pressure difference shrinks and the breeze weakens. Mirrors reflect light, not 'wind,' so they add no wind. The breeze certainly depends on land heating — that is its cause. And reflecting sunlight away cools the land relative to normal, it does not make the land hotter than the sea, so no reversal to a strong land breeze. Less land heating means a weaker sea breeze.

38. (D) In winter the land cools faster and more than the sea, so the cooler land develops higher pressure; wind blows from that higher pressure over the land toward the lower pressure over the relatively warmer sea — hence land to sea. Because this air originates over dry land and has not travelled across the sea collecting moisture, it is dry, so it brings little rain. The choice saying the sea cools faster reverses the land-sea cooling rates. Winter winds are roughly horizontal land-to-sea winds, not straight-up snow-bringers. And cold air does not 'always carry more water'; these land winds are dry. Cooler land giving higher pressure and dry origin explains both direction and dryness.

39. (A) A cyclone's centre, the eye, is calm and clear; the fiercest winds and heaviest rain form the eye wall, the ring just around the eye. So the most violent zone is not the geometric centre but the eye-wall ring, and the eye itself is comparatively safe — a position at the centre does not equal maximum violence. The 'eye is most violent' choice is the very misconception being corrected. The calm eye and violent eye wall are not equally dangerous — they differ sharply. And the outer cloud bands are milder than the eye wall, so calling the eye wall calmest is exactly backwards. The eye wall around the calm eye is the deadliest part.

40. (D) Cyclone, hurricane and typhoon are different regional names for the very same type of large tropical storm — the name depends on where in the world it occurs, not on size. So treating them as three distinct storms of fixed, ranked sizes is the error; none is permanently the largest. Re-ranking which is biggest still wrongly assumes fixed different sizes. They are not genuinely different storms. And a typhoon is not a tornado — a tornado is a separate, much smaller phenomenon. The real error is calling regional names of one storm three different fixed-size storms.

41. (B) Over the heated patch the warmed air expands, becomes less dense and rises away, so less air remains pressing on the surface and the surface pressure falls. The shaded patch is not heated this way, so its air stays put and its surface pressure stays comparatively high — this very difference drives a breeze toward the heated patch. The 'pressure rises because warm air is heavier' choice repeats the density error; warm air is lighter and rises. Heating one patch lowers its pressure, not both patches equally. And the pressure falls modestly as air rises away, it does not drop to zero — the air is not removed entirely. So the heated patch's surface pressure falls.

42. (C) Inside both jars the trapped air pushes out with the same pressure. At the valley the outside air pushes in about as hard, so the strains nearly balance. On the peak the outside air is thinner and pushes in less, while the inside push is unchanged, so the net effect is an outward strain on the walls. The 'inward squeeze' choice wrongly says mountains have stronger pressure — they have weaker pressure. The two jars do not feel the same strain, because the outside pressure differs with altitude. And air does not lose all its pressure on a mountain; it is merely thinner. Weaker outside push with unchanged inside push gives a net outward strain.

43. (A) Storing clean drinking water ahead of a cyclone is sound because floodwater can contaminate sources and the storm can cut off normal supply. Throwing open all doors and windows lets the wind and rain into the house and can help tear the roof off — it is unsafe. Going to the seashore during a cyclone warning exposes you to surge and violent waves — dangerous. And sheltering under a lone tall tree is hazardous, as it can be blown down or struck. Storing clean water in advance is the safe, correctly justified action.

44. (B) From the centre out, a cyclone has the calm, clear, low-pressure eye; then the eye wall, the ring of fiercest winds and heaviest rain; then the outer spiralling cloud bands. The first wrong choice mislabels the eye as violent and high-pressure (it is calm and low-pressure) and the eye wall as calm. The second again calls the eye high-pressure and puts the fiercest winds at the outer rim instead of the eye wall. And a cyclone certainly has a violent zone — the eye wall — so 'calm everywhere' is wrong. Calm low-pressure eye, violent eye wall, outer spiralling bands is correct.

45. (C) The cyclone's intensification runs on warm sea water supplying warm moist air that condenses and releases heat to deepen the central low. Take away the warm water — move it over cool water — and little warm moist air rises and little condensation heat is released, so the low stops deepening and the storm stalls. Adding more vapour, deepening the low further, or supplying more warm moist air all feed the cycle and would intensify, not stop, the storm. Removing the warm sea water is the change that starves the engine.

46. (D) In hot water the air expanded and swelled the balloon. Moving to ice water cools the air, which contracts; its pressure falls below the outside air pressure, so the higher outside pressure pushes the balloon inward and it collapses. Cold air contracts — it does not expand — so 'swells even further' is wrong. Cooling certainly does affect it (it reverses the swelling), so 'stays swollen' is wrong. And the collapse is driven by the inside-versus-outside pressure difference, not by the rubber pulling itself in. Contracting air with pressure below outside, pushed in by outside air, is the answer.

47. (D) The sea breeze needs the land warmer than the sea, giving lower pressure over land that draws sea air in. On the overcast day the land heats only a little, so the difference shrinks and the breeze weakens. On the cold-spell day the land is colder than the sea, so the pressure pattern flips and the surface wind tends to reverse into a land breeze. Cloud and cold do not by themselves increase wind. The breeze does depend on land temperature, so 'exactly the same' is wrong. And cold land air rushing out toward the warm sea is a land breeze, not a stronger sea breeze. So: weaker, then reversed.

48. (B) Falling pressure (barometer dropping fast) combined with strengthening wind at sea is the classic sign of an approaching low-pressure system or storm, so the responsible advisory is to warn of it and bring the boats in. Falling pressure does not promise settled fair weather — the opposite. The combined readings are far from meaningless; they are a clear warning, so telling the fishermen to keep fishing far out would be dangerous. And a

fast-falling barometer is not a forecast of a freeze; it is about an approaching storm. Warning of an approaching storm and recalling the boats is the safe call.

49. (D) The fast-moving air in the stream is at lower pressure than the slower surrounding air, so whenever the ball drifts to the edge, the higher outside pressure pushes it back into the stream — and this works along whatever direction the stream points, including a 45-degree tilt. The ball is not weightless; it has weight that the stream supports while the pressure difference centres it. The air is not 'sticky glue' holding it on a path — that is a vague non-mechanism. And there is no magnetism. The lower pressure inside the fast stream, with higher pressure around it pushing inward, keeps the ball in the slanted stream.

50. (A) First the Sun warms the ground and the surface air on a hot humid day (iv). That warm moist air rises and cools as it goes up (ii). On cooling, its water vapour condenses into towering tall clouds (iii). As the cloud droplets merge and grow too heavy to stay aloft, they fall as rain (i). So the order is iv, ii, iii, i. Starting with rain falling (i first) puts the result before its cause. Starting with clouds already formed (iii first) skips the warming and rising that make them. And starting with air already rising (ii first) leaves out the initial warming that lifts it. Warming, rising, condensing, raining is the true sequence.

51. (B) A narrow whirling column only a few hundred metres wide, reaching from a cloud to the ground, short-lived and cutting a narrow destructive track, is a tornado. A tropical cyclone is hundreds of kilometres across and lasts far longer — far bigger than this. A sea breeze is a gentle horizontal coastal wind, not a violent vertical column. And a high-pressure region is a broad calm system, the opposite of an intense narrow whirl. These features identify a tornado.

52. (C) The single underlying fact is that land heats and cools faster than water. By day the land heats quickly, becomes warmer than the sea, develops lower pressure, and draws cool sea air inland — a sea breeze. By night the land cools quickly, becomes cooler than the sea, so now the warmer sea air rises and air flows out from the land — a land breeze. The Sun does not 'blow' wind directly; it heats surfaces unevenly. The sea does not change temperature faster — it is the slower one, which is the whole point. And the reversal is not a fixed clockwork; it is driven by the changing land-sea temperature difference. Land's faster temperature change underlies both breezes.

53. (A) Wind speed in a cyclone is driven by the pressure difference between the surroundings and the centre. A deeper central low means a bigger difference, so the surrounding higher-pressure air rushes inward faster — stronger winds. A deep low does not still the air; it accelerates it inward. Cyclones draw air inward, not push it outward. And the central pressure is closely tied to wind speed (sea temperature matters by feeding the low, but the immediate driver of wind is the pressure difference). A deeper low gives a bigger difference and faster inrushing wind.

54. (B) Fast wind over a roof lowers the pressure above it while the still air inside stays at higher pressure, so the roof is pushed up; the best defence is a heavy, well-anchored, low-profile roof, which is firmly fixed down and presents less surface to the lift. A light, loosely laid roof is the easiest to lift, not the safest. A tall, broad, unanchored roof offers a large surface for the lift and is not held down by size alone. And fast wind over a roof lowers the pressure to lift it, not press it down. A heavy, anchored, low roof resists the lift best.

55. (B) The flame stays bright because the chimney sets up a draught: warm air rises out the top and fresh air is drawn in at the bottom, supplying the flame. Sealing the top stops the warm air escaping, so fresh air no longer flows in, the draught fails, and the flame dims or goes out. Trapping the warm air does not feed the flame — it suffocates it. The flame does depend on the through-flow of air, so 'unaffected' is wrong. And blocking the top stops the draught rather than speeding it. Sealing the top kills the draught, so the flame dies.

56. (D) The cover is held up only when the upward air pressure on it exceeds the total downward force, which is the water's weight plus the cover's own weight. A light card adds almost nothing, so it holds. A heavy metal plate adds a large downward weight, so the total downward force may now exceed the upward air pressure and the plate falls. 'Heavier covers gripped harder' invents a non-existent grip. The cover's weight does matter — it adds to the downward side. And air pressure pushes up on the cover but does not lift heavy metal away upward. The plate's extra weight can overcome the air pressure, so it may fall.

57. (C) A cyclone is fed by warm moist air rising off a warm sea; the condensation of that vapour releases heat that deepens its low and powers its winds. Once over land, it loses that supply of warm moist air, so the cycle is starved and the storm weakens within a day. The storm does not 'freeze solid' over land. Buildings do not form a wall that blocks all the wind across a storm hundreds of kilometres wide. And the eye does not snap shut on contact with soil. Being cut off from warm, moist sea air is the correct reason for the weakening.

58. (D) Faster eye-wall winds go with a deeper, lower central pressure, since the bigger pressure difference between the surroundings and the centre drives faster inrushing wind. M's winds (200 km/h) are faster than N's (120 km/h), so M most likely has the lower (deeper) central pressure. The choice giving N the lower pressure inverts the relationship. Forming over the same sea on the same day does not force identical central pressures — their wind speeds already differ. And higher central pressure does not 'fling winds outward faster'; cyclonic winds rush inward, driven by a deeper low. So M has the deeper low.

59. (A) Both seasonal reversals trace to one fact: land heats and cools faster than the sea. In summer the land becomes much hotter than the sea, forming lower pressure that pulls moist sea air inland (the wet monsoon). In winter the land cools faster and becomes cooler than the sea, forming higher pressure that pushes dry air out to sea. The Moon's gravity governs tides, not these seasonal winds. The sea does not change temperature faster — it is

the slower one, which is exactly why the land's swings drive the reversal. And the monsoon is not a calendar reflex divorced from heating; it is driven by the land-sea temperature difference. Land's faster temperature change underlies both.

60. (D) As a cyclone's centre passes over a town, the leading eye wall arrives first with fierce wind from one direction (ii). Then the calm, clear eye passes over (i). Then the trailing eye wall brings fierce wind from the opposite direction (iii). Finally the outer bands move off and the wind and rain ease (iv). So the order is ii, i, iii, iv. Starting with the calm (i first) puts the eye before the leading eye wall. Putting both fierce winds back to back with no calm between (ii then iii) skips the eye passing over. And the easing-off (iv) is the end, not the start. Leading eye wall, calm eye, trailing eye wall, then easing is the true sequence.